CodePilot

AI-Assisted ICD-10-CA Coding — Technical Build & Deployment
Full Stack Data Science Systems
Group 7

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Agenda

Problem

Manually matching a clinical note to the correct diagnosis code is slow and error-prone.

- The ICD-10-CA catalog (Ontario Health / CIHI) contains 15,573 codes — too many to search manually per chart.
- Coding accuracy directly affects Ontario's HBAM funding model, not just billing — errors have real financial consequences.
- The medical coding workforce is shrinking and aging, increasing per-chart turnaround time.

Why This Problem

30%	50+	15,573
national coder shortage	average coder age	ICD-10-CA codes to search

This is a real, present-tense operational problem — not a hypothetical future one. It affects funding accuracy, staff workload, and turnaround time today.

Literature Review

Past Available Projects

- Nuance / Microsoft DAX — ambient documentation + coding
- 3M 360 Encompass — enterprise coding & CDI platform
- Fathom Health, CodaMetrix — autonomous AI coding startups

Similar Academic Approaches

- CNN/BERT-based ICD classifiers trained on MIMIC-III/IV
- ClinicalBERT for readmission & coding prediction
- Multi-label classification with 40K+ labeled notes

Drawbacks We Identified

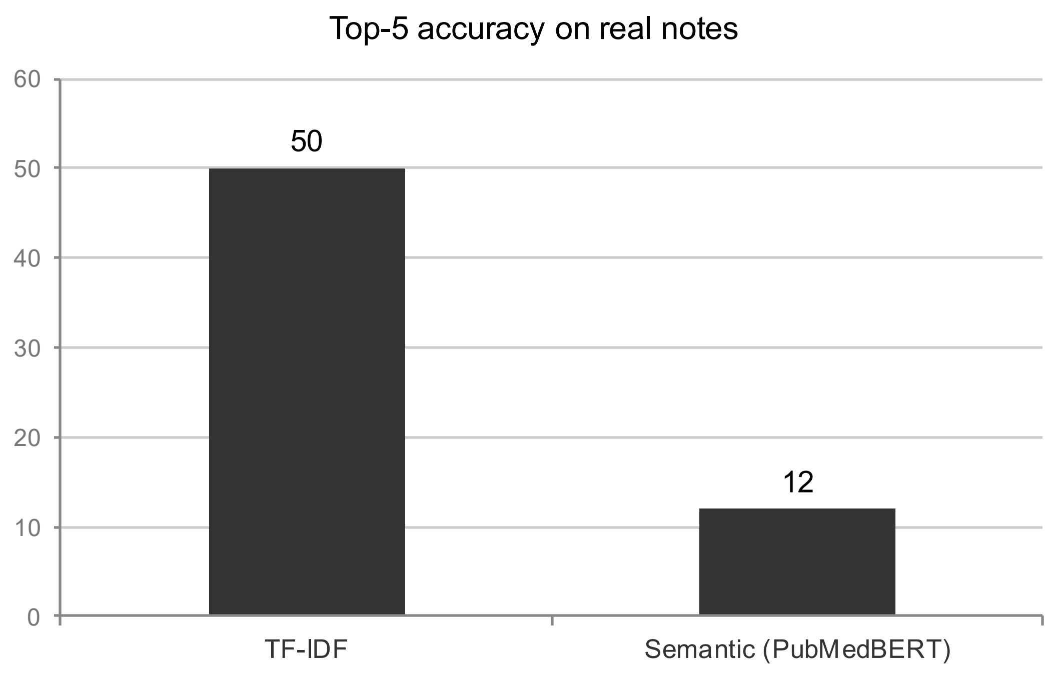
- Enterprise platforms priced for large multi-site health systems, not a single mid-size hospital
- Most academic work trains on MIMIC (ICD-9 or US ICD-10-CM) — not Canada's ICD-10-CA classification
- Classifier-based approaches need thousands of labeled (note, code) pairs — infeasible for us to assemble at project timeline

ML Canvas

Value Proposition Coders get top-5 ranked ICD-10-CA suggestions per note instead of searching 15,573 codes manually	**Data Sources** Official ICD-10-CA catalog (Ontario Health/CIHI) + 16 hand-verified real notes (MTSamples, CC0)	**Prediction Task** Ranked retrieval: note text -> top-k most similar code descriptions
Decisions Coder confirms, overrides, or requests more candidates — model never auto-submits	**Building Models** TF-IDF baseline + pretrained PubMedBERT embeddings — no training from scratch	**Offline Evaluation** Top-1/Top-5/category-match accuracy + latency, benchmarked on real notes
Features Free-text note only (current scope) — no structured EHR fields yet	**Live Evaluation (planned)** Top-5 acceptance rate, override rate, time-per-chart vs. baseline	**Risks** Multi-comorbidity notes, semantic model catalog bias — both tested & documented

Model Benchmarking

Two pretrained approaches, benchmarked on 16 real, hand-verified clinical notes against the ICD-10-CA catalog



Metric	TF-IDF	Semantic
Top-1 accuracy	12%	0%
Top-5 accuracy	50%	12%
Category match	25%	25%
Avg latency	0.75 ms	17.3 ms

Key finding: the simpler keyword model outperformed the pretrained semantic model — root-caused to catalog composition (9.4% of ICD-10-CA is obstetric codes with repetitive phrasing), not just "the model is worse."

Model Deployment

Notebook

Initial exploration:
TF-IDF + semantic
model prototyping

→

Script

model_baseline.py
model_semantic.py
(standalone, testable)

→

Flask API

api.py
/health /predict
endpoints

→

Docker

Dockerfile
built & tested
locally

→

Cloud Host

Render (free tier)
live public URL
+ demo webpage

Live: https://icd10-coder.onrender.com

Source: https://github.com/Vikashkhatri/icd10-coder

Challenges & Issues Encountered

Apple Silicon numerical bug

Semantic model produced NaN/Inf values on MPS backend — fixed by forcing CPU + sanitizing embeddings

MIMIC-III demo has no notes

PhysioNet strips all clinical text from the demo tier by design — pivoted to public MTSamples corpus

Wrong code system initially

Prototype started on US ICD-10-CM; corrected to the real ICD-10-CA standard used by Ontario hospitals

Semantic model catalog bias

Unexplained drift toward obstetric codes — root-caused to catalog composition, not a simple bug

Multi-comorbidity notes

Neither whole-note nor sentence-split querying reliably handles 3+ conditions in one note — documented as an open limitation

No 500+ real labeled dataset

Investigated MIMIC-IV, Kaggle, CodiEsp — concluded credentialed access is the only trustworthy path at scale

Demo

Live, public, deployed right now

icd10-coder.onrender.com

Clinical note:

"45 year old male with type 2 diabetes mellitus, poorly controlled, presenting for routine follow-up."

Suggested ICD-10-CA codes:

E11.33	Type 2 diabetes with other retinopathy
E10.33	Type 1 diabetes with other retinopathy
E11.41	Type 2 diabetes with polyneuropathy

Model: TF-IDF · Catalog: ICD-10-CA (15,573 codes) · Deployed on Render (free tier)

Q & A

github.com/Vikashkhatri/icd10-coder · icd10-coder.onrender.com